NATIONAL UNIVERSITY OF SINGAPORE
Department of Statistics and Data Science

ST5202: Regression Analysis

TUTORIAL 9

1. The `riboflavin` data contain gene expression measurements for 71 samples, with 4088 gene expression levels recorded for each sample. For each sample, the log-transformed riboflavin production rate is also recorded. In this tutorial, use `y` as the response variable and the 4088 gene expression levels in `x` as predictor variables.

   Use the following R code to download and prepare the data.

```
library(hdi)
library(glmnet)
library(ncvreg)

data(riboflavin)

X <- as.matrix(unclass(riboflavin$x))
y <- as.numeric(riboflavin$y)

set.seed(123)
n <- length(y)
train_id <- sample(1:n, round(0.7*n))
test_id  <- setdiff(1:n, train_id)

X_train <- X[train_id, ]
X_test  <- X[test_id, ]
y_train <- y[train_id]
y_test  <- y[test_id]
```

   Now, we have two datasets: `X_train` and `y_train` is the training data we will use to train our model (including the tuning parameter selection); and `X_test` and `y_test` is the test data where we will compare models.

   For cross-validation, use the following fold assignment so that your results are reproducible.

```
set.seed(123)
foldid <- sample(rep(1:10, length.out = length(y_train)))
```

(i) Fit a LASSO model to the training data by 10-fold cross-validation, where the cross-validation divisions are generated in the main problem as `foldid`.

Use the following code as a template.

```
cv_lasso <- cv.glmnet(X_train, y_train,
                      alpha = 1,
                      standardize = TRUE,
                      foldid = foldid)
plot(cv_lasso)
cv_lasso$lambda.min
```

Draw the cross-validation plot. Report the selected value of λ and the number of nonzero coefficients. Apply the fitted model to `X_test`, and report the mean squared error between the fitted values and `y_test`.

The procedures are carried out by the following R codes:

```
cv_lasso <- cv.glmnet(X_train, y_train,
                      alpha = 1,
                      standardize = TRUE,
                      foldid = foldid)
plot(cv_lasso)

lasso_lambda <- cv_lasso$lambda.min
lasso_lambda

lasso_fit <- glmnet(X_train, y_train,
                    alpha = 1,
                    standardize = TRUE,
                    lambda = lasso_lambda)
lasso_pred <- as.numeric(predict(lasso_fit, newx = X_test))
lasso_mse <- mean((y_test - lasso_pred)^2)
lasso_mse

lasso_coef <- as.matrix(coef(lasso_fit))
lasso_nz <- sum(lasso_coef[-1,1] != 0)
lasso_nz
```

The selected value of λ is

0.09573870

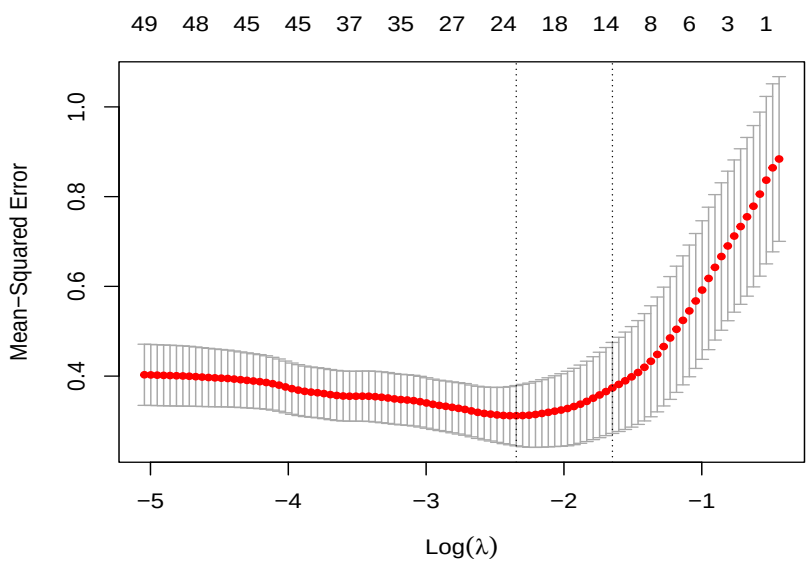
The number of nonzero coefficients is

23

The test MSE is

0.2066601

The cross-validation plot is shown below.



(ii) For MCP and SCAD, repeat part (i).
Use the following code as a template.

```
cv_mcp  <- cv.ncvreg(X_train, y_train,
                     penalty = "MCP", fold = foldid)
cv_scad <- cv.ncvreg(X_train, y_train,
                     penalty = "SCAD", fold = foldid)

plot(cv_mcp)
plot(cv_scad)

mcp_coef  <- coef(cv_mcp,  lambda = cv_mcp$lambda.min)
scad_coef <- coef(cv_scad, lambda = cv_scad$lambda.min)
```

The procedures are carried out by the following R codes:

```
cv_mcp <- cv.ncvreg(X_train, y_train,
                    penalty = "MCP", fold = foldid)
plot(cv_mcp)

mcp_lambda <- cv_mcp$lambda.min
mcp_lambda
mcp_pred <- as.numeric(predict(cv_mcp$fit, X_test,
                               lambda = mcp_lambda))
mcp_mse <- mean((y_test - mcp_pred)^2)
mcp_mse
mcp_coef <- coef(cv_mcp$fit, lambda = mcp_lambda)
mcp_nz <- sum(mcp_coef[-1] != 0)
mcp_nz

cv_scad <- cv.ncvreg(X_train, y_train,
                     penalty = "SCAD", fold = foldid)
plot(cv_scad)

scad_lambda <- cv_scad$lambda.min
scad_lambda
scad_pred <- as.numeric(predict(cv_scad$fit, X_test,
                                lambda = scad_lambda))
scad_mse <- mean((y_test - scad_pred)^2)
scad_mse
scad_coef <- coef(cv_scad$fit, lambda = scad_lambda)
scad_nz <- sum(scad_coef[-1] != 0)
scad_nz
```

For MCP, the selected value of λ is

```
0.11842249
```

the number of nonzero coefficients is

```
6
```

and the test MSE is

```
0.4399634
```

For SCAD, the selected value of λ is

`0.07297376`

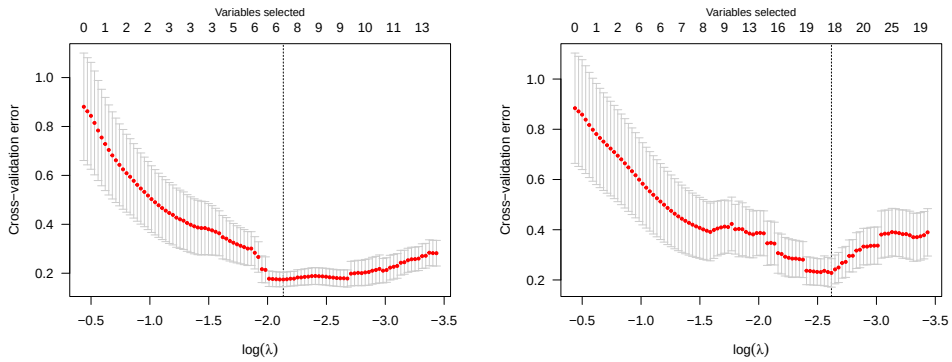
the number of nonzero coefficients is

`17`

and the test MSE is

`0.3669282`

The cross-validation plots for MCP and SCAD are shown below.



(iii) Present your results for lasso, MCP and SCAD in a table. The table should include the selected tuning parameter, the number of nonzero coefficients, and the test MSE.

Based on your results, which method gives the smallest test MSE? Which method gives the sparsest model? Give some comments.

The results are summarized in the following table:

Method	*Selected* λ	*Nonzero coefficients*	*Test MSE*
Lasso	*0.09573870*	*23*	*0.2066601*
MCP	*0.11842249*	*6*	*0.4399634*
SCAD	*0.07297376*	*17*	*0.3669282*

The method with the smallest test MSE is `Lasso`*. The sparsest model is given by* `MCP`*.*

In this example, lasso gives the best prediction performance on the test set, although it selects more predictors than MCP and SCAD. This suggests that the stronger shrinkage in lasso leads to a better bias–variance trade-off for prediction. MCP gives the sparsest model, selecting only 6 predictors, but its test MSE is the largest. SCAD is in between lasso and MCP in terms of both sparsity and prediction error.

Therefore, in this example, lasso is preferred if the main goal is prediction, while MCP is preferred if the main goal is to obtain a sparse model.

2. For the **SMSA data**, consider the `number of active physicians` as the response variable and the others as predictor variables.

   (i) Carry out the forward selection procedures for the model selection, taking the lower model as the one containing the factor variable `geographic region` and the upper model as the full model containing all the predictor variables. Is the model selected the same as that selected by the forward selection procedure with the lower model as the null model? If not, select among the two models by using AIC.

   The procedures are carried out by the R codes:

```
SMSA.dat = read.table("D://Rsession/SMSA.txt",header=TRUE)

y=SMSA.dat$doctor; x1=SMSA.dat$Land; x2=SMSA.dat$T.p; x3=SMSA.dat$P.city;
x4=SMSA.dat$p.65; x5=SMSA.dat$beds; x6=SMSA.dat$h.sch;x7=SMSA.dat$labor;
x8=SMSA.dat$income; x9=SMSA.dat$crimes;
x10=factor(SMSA.dat$region);
p.dat=data.frame(y,x1,x2,x3,x4,x5,x6,x7,x8,x9,x10)


## (i) selecting models
library(MASS)

# Starting models for the selection procedure:
null1 = lm(y~1, data=p.dat); null2 = lm(y~x10, data=p.dat)

# Selection with null model as starting model
f1=stepAIC(null1,scope=list(upper=~x1+x2+x3+x4+x5+x6+x7+x8+x9+x10,lower=~1),
           direction="forward")
summary(f1)

# Selection with starting model containg x10
f2=stepAIC(null2,scope=list(upper=~x1+x2+x3+x4+x5+x6+x7+x8+x9+x10,lower=~x10),
           direction="forward")
summary(f2)
```

   The model selected while taking null model as the starting model is

   `y ~ x8 + x9 + x5 + x2 + x10 + x1`

   The model selected while taking `y~x10` *as the starting model is*

```
y ~ x10 + x9 + x5 + x1 + x6
```

The AIC values of the two models are:

```
  AIC(f1)   AIC(f2)
 2127.556  2132.503
```

The model selected while taking null model as the starting model is better and is finally selected.

(ii) For the model selected in (i),

The raw materials used for the following parts are computed by the R codes:

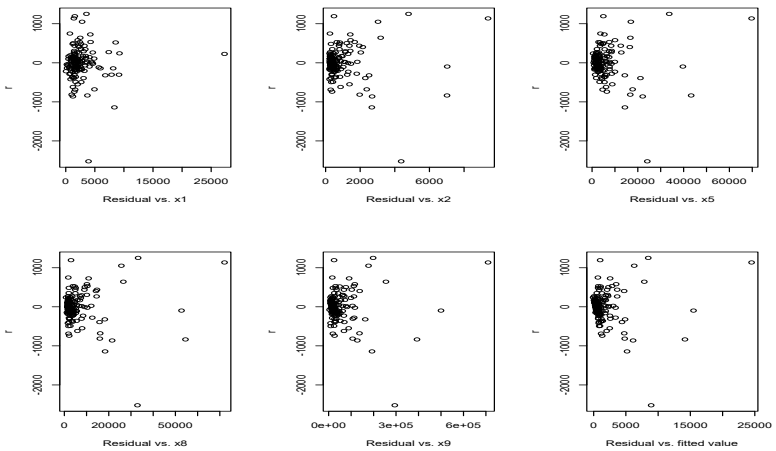
```
fit1=glm(y~x1 + x2 + x5 + x8 + x9 + x10,data=p.dat)
yhat = fit1$fitted.values
r = residuals(fit1,type="pearson")
h = hatvalues(fit1,type="diagonal")
infl = influence(fit1, do.coef = FALSE)
rsta = rstandard(fit1, infl, type = "pearson")
rstu = rstudent(fit1, infl, type = "pearson")
cook =cooks.distance(fit1, infl,res = infl$pear.res,
dispersion = summary(fit1)$dispersion,hat = infl$hat)
```

(iia) Check whether or not the regression function is linear in terms of the numerical predictor variables.

The linearity is checked by plotting the Pearson's residual against each of the predictors, which are carried out by the R codes:

```
par(mfrow=c(2,3))
plot(x1, r,xlab="Residual vs. x1")
plot(x2, r,xlab="Residual vs. x2")
plot(x5, r,xlab="Residual vs. x5")
plot(x8, r,xlab="Residual vs. x8")
plot(x9, r,xlab="Residual vs. x9")
plot(yhat,r,xlab="Residual vs. fitted value")
```

The plots are as follows:

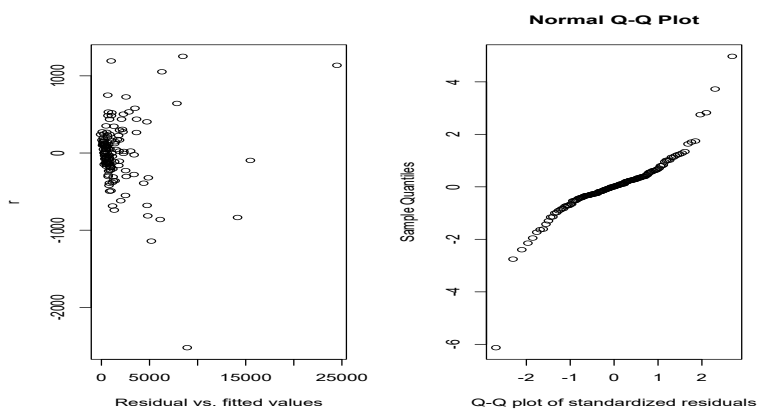


Since there is no obvious trend showing up in the plots, it can be considered that the linearity assumption holds.

(iib) Check whether or not the assumption of constant variance is violated.

(iic) Check whether or not the assumption of normality is violated.

The constancy of variance is checked by the plot of Pearson's residual against the fitted values. The normality is checked by the normal probability plot of the studentized residuals. The plots are shown below:



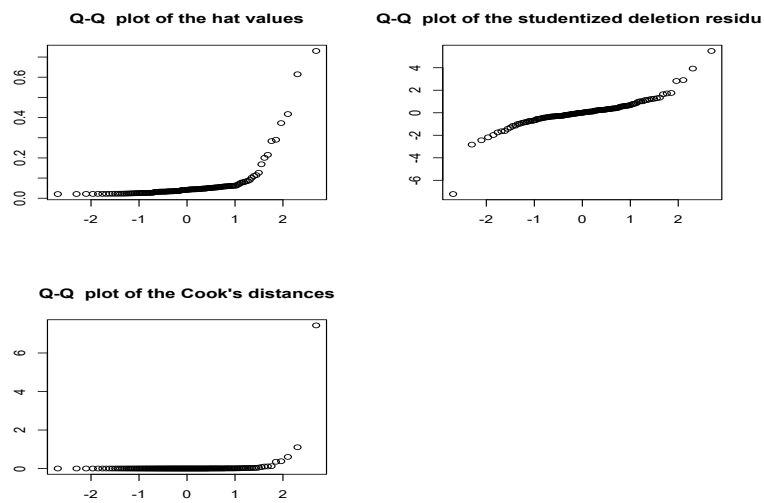
The plot of Pearson's residuals against the fitted values does not show obvious evidence for non-constancy of variance, but it reveals that there are some outliers. The normal probability plot suggests that the normality is violated and the distribution of the data might be a symmetric distribution with heavier tails than

the normal distribution.

(iii) Compute the hat values, studentized deleted residuals and Cook's distances.

(iiia) Draw the Q-Q plot for each of these measures.

The plots are given below:



(iiib) Identify the observations whose hat values exceeds $2p/n$, where p is the number of parameters in the regression function including the dummy variables for the factor `geographic region`.

$n = 141, p = 9, 2p/n = 0.1276596$
The observations are identified by `h[h>=2*p/n]` *as*

```
     1      2      3      4      6      7      8     12     27
 0.730  0.290  0.373  0.417  0.200  0.283  0.168  0.215  0.615
```

(iiic) Identify the observations whose studentized deleted residual has a absolute value bigger than $2d$ where d is the sample standard deviation of the studentized deleted residuals.

The sample standard deviation of the residuals is obtained by `sqrt(var(rstu))` *as* 1.179096, $2d = 2.358$. *Th observations are identified by* `rstu[abs(rstu)>2*d]` *as*

```
    1      3      4      5      7      9     68
5.490 -2.440  3.922 -7.216  2.900 -2.828  2.817
```

(iiid) Identify the observations whose Cook's distance is bigger than 0.1.

The observations are identified by `cook[cook>0.1]` *as*

```
    1     3     4     5     7     8     9    27
7.431 0.379 1.104 0.603 0.349 0.104 0.109 0.122
```

(iv) Conduct formal tests by creating dummy variables to check simultaneously whether or not the observations identified in (iiid) are outliers, using Bonferroni's criterion to control the overall type I error rate at level 0.05.

The dummy variables are created by the codes:

```
u1=u2=u3=u4=u5=u6=u7=u8=rep(0,n)
u1[1]=1; u2[3]=1; u3[4]=1;u4[5]=1; u5[7]=1; u6[8]=1; u7[9]=1; u8[27]=1
```

The model is re-fitted to the original variables together with the dummy variables:

```
fit2=glm(y~x1 + x2 + x5 + x8 + x9 + x10 +u1+u2+u3+u4+u5+u6+u7+u8)
summary(fit2)
```

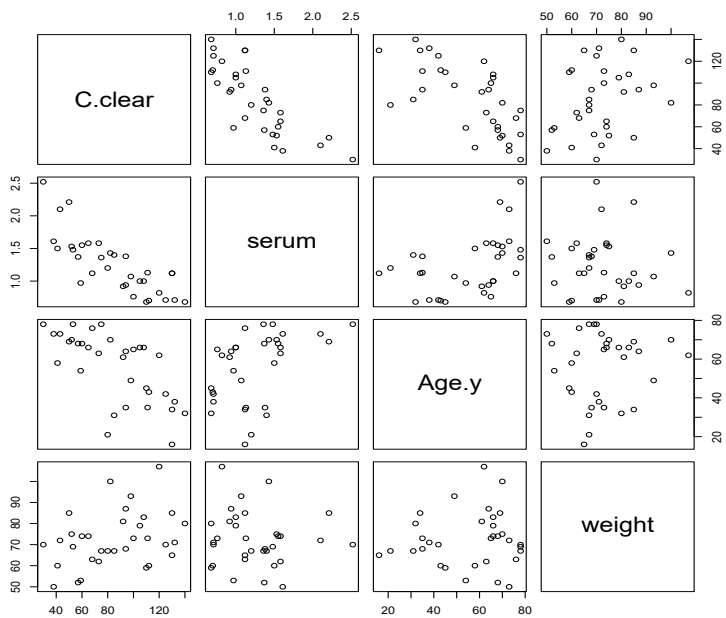
The summary results are:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.894e+01  7.459e+01  -1.327  0.18710
x1          -7.538e-03  1.675e-02  -0.450  0.65350
x2          -9.152e-01  3.994e-01  -2.292  0.02362 *
x5           7.897e-02  1.922e-02   4.109 7.18e-05 ***
x8           2.310e-01  5.140e-02   4.493 1.59e-05 ***
x9           1.346e-02  2.766e-03   4.866 3.38e-06 ***
x102        -3.286e+01  7.855e+01  -0.418  0.67644
x103         5.540e+01  8.119e+01   0.682  0.49633
x104         1.109e+02  1.048e+02   1.058  0.29208
u1           2.628e+03  5.669e+02   4.636 8.88e-06 ***
u2          -1.399e+03  4.375e+02  -3.198  0.00176 **
u3           1.236e+03  4.103e+02   3.013  0.00314 **
u4          -2.909e+03  3.162e+02  -9.198 1.10e-15 ***
u5           5.288e+02  3.750e+02   1.410  0.16095
u6          -6.129e+02  3.406e+02  -1.800  0.07436 .
u7          -1.370e+03  3.128e+02  -4.381 2.49e-05 ***
u8          -1.375e+02  4.700e+02  -0.292  0.77043
```

The Bonferroni critical values is obtained by `qt(0.05/(2*8),124,lower.tail=FALSE)` *as* 2.781854. *By controlling the overall type I error rate at 0.05, it can be claimed that the observations 1 (u1), 3 (u2), 4(u3), 5(u4), 9(u7) are influential outliers.*

3. Creatinine clearance is an improrrtant measure of kidney function but is difficult to obtain in clinical office setting because it requires 24 hour urine collection. To determine whether this measure can be predicted from some data that are easily available, a kidney specialist obtained the data for 33 males. The data consists of creatinine clearance (Y), serum creatinine concentration (X_1), age (X_2) and weight (X_3). The data is provided in the file `kidney.txt` on Canvas.

(i) The following is the pairwise scatter plots:

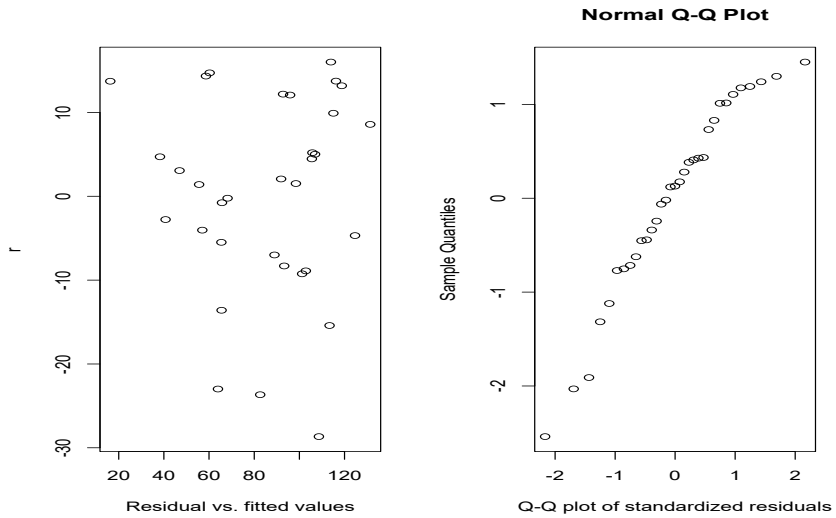


From the plots what relations between the response variable `C.clear` and each of the covariates you can suggest? Provide your reasons.

There are a slight curvature in the plots of `C.clear` *(Y) against* `serum` *(X_1) and against* `Age.y` *(X_2), which suggests a non-linear relationship between Y and X_1, and between Y and X_2. A possible scale for X_1 could be $1/X_1$, for X_2 could be $\sqrt{X_2}$. The plot of Y against X_3 shows a linear trend and there is no obvious curvature, it can be considered that the relation is linear.*

(ii) The model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$ is fitted to the data.

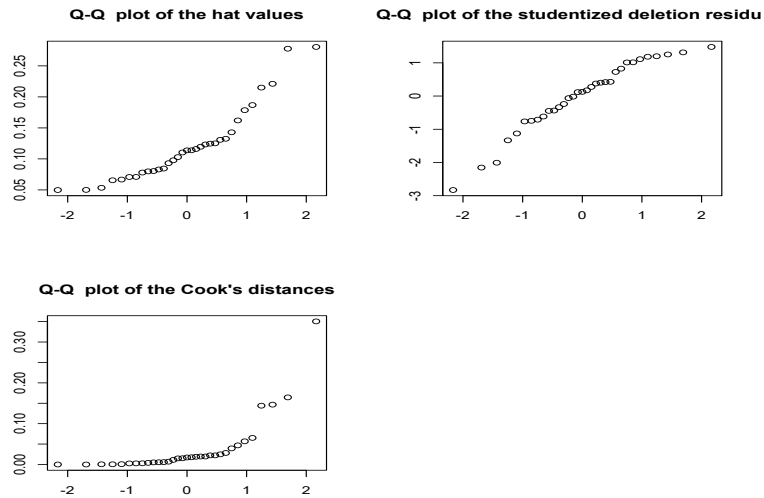
(iia) The following are the plot of residuals vs. fitted values (left panel) and the Q-Q plot of the studentized residuals (right panel):



Are the plots "null patterns"? If they are not null patterns, what could be the possible discrepancies? Explain.

Both plots are not the null pattern. The possible discrepancies indicated by the plot of residuals vs. fitted values could be either the variance is not constant or the regression function is not appropriate.
When we draw the line $y = x$ in the second plot, it can be found that there are a few points which fall below the line. It might be caused either by non-normality or by some outliers.

(iib) The following are the Q-Q plots of the three outlier measures: hat value, studentized deletion residual and Cook's distance.



Circle out the possible outliers in terms of the three measures.

(iic) A formal test is to be carried out to check formally whether or not the observations 16,20,21,26 and 29 are significant outliers. The following are the R codes and the fitted results:

```
u1=rep(0,n);u2=u1;u3=u1;u4=u5=u1;
u1[16]=1;u2[20]=1;u3[21]=1;u4[26]=1; u5[29]=1
outlier=glm(C.clear~serum+Age.y+weight+u1+u2+u3+u4+u5,
           data=kidney.dat)
summary(outlier)
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 143.3637    11.8931  12.054 1.14e-11 ***
serum       -42.4945     4.6435  -9.151 2.70e-09 ***
Age.y        -0.8176     0.1218  -6.714 6.05e-07 ***
weight        0.5890     0.1281   4.597 0.000116 ***
u1           16.2631    10.3110   1.577 0.127827
u2            9.0249    10.1591   0.888 0.383165
u3          -30.2122     9.3382  -3.235 0.003525 **
u4          -34.6656     9.9302  -3.491 0.001884 **
u5          -26.5429     9.0991  -2.917 0.007551 **
```

Using Bonferroni approach to control the overall type I error rate at level $\alpha = 0.05$, which of the five observations are significant outliers?

Multiplying the individual p-values for each of the dummy variables by 5, the Bonferroni p-values for $u3, u4, u5$ *are less than 0.05. Therefore, observations 21, 26 and 29 can be claimed as significant outliers.*